Literature Review & Elaboration of Problem

Physical Layer Security in 5G Communication Pipeline

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1.1(Intial pipeline (Ayan & Rayan))	21st Sept'25	Demo OFDM pipeline for Proposal Defense
1.2 (Structured two-way functional pipeline Rayan, Mujtaba))	10th Oct'25	Sender Receiver instance implementation along with base stations
1.3 (Setting up Agent 1 Ayan Mujtaba))	20th Oct'25	Implemented Agent for Resource Allocation
1.4 (Setting up Agent 2 Ayan, Rayan)	28th Oct'25	Implement Agent For subcarrier selection
1.5 (visual representations Mujtaba)	15th Nov'25	Fixed and added all the graphical plots
1.6 (Spectrum Sensing Ayan Rayan Mujtaba))	28th Nov'25	Introduce and implemented the spectrum sensing

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1.1	Dr. Farrukh Shahid	Supervisor	20th Sep’25
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1. Abstract

The explosion of connected devices caused severe spectrum shortage, as static spectrum allocation leads to underutilization and spectrum holes [12], [13]. For this reason, Cognitive Radio (CR) and Dynamic spectrum access (DSA) technologies enable unlicensed Secondary users to opportunistically Access Spectrum bands licensed to Primary users [12], [13]. Based on that need, this project is aiming at designing and implementing an energy-aware Cognitive Radio Network (CRN) framework. It is designed so that Primary Users are given strict priority and Secondary Users get maximum throughput. To this end it couples a data-driven physical (PHY) layer whose signal energy comes directly from actual QAM-modulated message bits [9] and a central medium access Control (MAC) layer controlling network Access via a Base Station coordinator.

A methodology was followed starting with socket-based network architecture. It's a client-server topology with socket programming, where Base Stations relay traffic between sending and receiving nodes. High fidelity modelling was done at the physical layer using a NumPy built simulation pipeline. This pipeline builds 16, 64 and 256-QAM constellations and Orthogonal Frequency Division Multiplexing (OFDM) signals with Cyclic Prefixes directly from text payload to calculate energy footprints [9]. Manage access the framework uses shared-state spectrum sensing via a Listen Before-Talk (LBT) protocol [14]. In this setup a local file is updated by the Base Station indicating channel state which Secondary senders read for BUSY states before transmitting, enabling a backoff when the channel is occupied [12], [13].

Also, priority queueing and traffic policing are handled via a retry queue at the Base Station. When the system detects Primary User IDs, a protection window opens; Incoming Secondary traffic is buffered instead of dropped during this time to ensure priority compliance. Security & session management are built into the design with AES GCM encryption [11] and Reed-Solomon (RS) error correction for data integrity. A custom handshake protocol was also implemented for persistent chat sessions between nodes. Also, the system implements counter-jamming to prevent attacks by secondary users by analysing graphs maintained in the backend node logs [2], [3], [4].

This research presents a bit level simulation environment for demonstrating the trade-offs between spectral efficiency and interference protection. Visualizing exact Energy Timeline and Constellation Diagrams of transmitted packets demonstrates secure, priority-based coexistence in software-defined environments [1], [7], [12].

2. Background and Justification

Radio spectrum normally belongs to fixed license holders Primary Users [1], [7]. Yet utilization differs significantly [12], [13]. Cognitive Radio enables Secondary Users to sense the environment and transmit during idle times [12], [13]. The message content must be decoupled from the signal energy in existing simulations; most simulators represent interference with random numbers [3], [4].

This research enhances the existing body of work by implementing a Data-Driven PHY-MAC Simulation:

1. Bit-Level Energy Computation: Instead of using standard simulations, this project calculates the Average OFDM Energy (np.mean(np.abs(ofdm_sig)**2)) by reconstructing the waveform from the actual message bits. That allows observing the effect of different payloads and Modulation Orders (M) on signal power [9], [10].
2. Cryptographic Spectrum Validation: The Base Station does not trust signal energy blindly. It tries to aes_gcm_decrypt and rs.decode the payload. Hence, security coupled with spectrum management justifies the research since a high energy signal failing decryption is considered noise or attack and only legitimate Primary Users are notified of the protection window [11].
3. Infrastructure-Assisted LBT: The simulation solves the Hidden Node Problem using the Base Station as central authority. The BS updates a global channel state file channel_state.json that sets the Busy/Idle status of all listening nodes for synchronized spectrum access [14].
4. Visual Analytics: Using matplotlib in combination to produce PDF reports and Energy Timelines to verify the modulation schemes and the Primary Protection Window implementation is an immediate visual confirmation of the modulation schemes and the enforcement of the Primary Protection Window [9].

3. Problem Statement

3.1. The Core Issue: Spectrum Scarcity vs. Static Allocation
Even as wireless data demands are rising, static spectrum allocation leaves licensed bands (Primary Users) idle for long periods. This inefficiency uses RF resources [1], [7].

3.2. Limitations of Abstract Simulations
One problem this code addresses is the lack of fidelity in standard CR simulations. Tools treat Signal Energy as an abstract variable. This project deals with coupling application layer data with physical layer attributes: a short text message using 16-QAM has a different spectral footprint than a large image file using 256-QAM. Most existing simulations miss this nuance [9].

3.3. The Priority Challenge in Shared Networks
In a shared network, Primary Users such as Emergency Services or Telecom Operators can not be guaranteed absolute priority. Secondary Users such as IoT sensors can collide without strict policing. This project tackles this by using Primary Protection Window logic: Once a Primary User transmits, the channel must be blocked for Secondary users for a specific duration (10 seconds) to ensure channel availability for subsequent Primary packets.

3.4. Session Management in Ad-Hoc Environments
Simulation difficulty anticipated and solved is persistent connections management in a connectionless UDP/TCP stream. The project solves the problem of interleaved messages by using a custom application layer handshake (CONNECT_REQUEST) where nodes create a dedicated Chat Partner state before exchanging data.

4. Literature Review

4.1. Spectrum Sensing Techniques

Literature classifies sensing into Energy Detection, Matched Filter, and Feature Detection [12], [13].

- Current Approach: This project uses hybrid Energy Detection technique. The Sender calculates the energy of its own OFDM frames and places this metadata in a Sensing Header. The Base Station confirms this by inspecting the frame. This differs from literature which employs Gaussian noise models. here the noise and signal are built from real text strings [10].

4.2. Proposed Methodology

The methodology adopts a cross-layer simulation using Python socket, threading, and numpy libraries:

- PHY Layer Construction: The Modulation Order (M) of the system is determined dynamically based on the message content (16 for short text, 256 for images/long text). It maps bits to complex symbols with QAM logic [9]

- I + j Q These are processed as Inverse Fast Fourier Transform (IFFT) to produce OFDM symbols with Cyclic Prefix (CP) to simulate realistic 5G waveforms [9]

- MAC Layer and Traffic Policing - The BS Logic: Base Station maps Node IDs to Operators:

  - S1-S50 (Primary): Jazz, WARID, UFONE, TELENOR & ZONG.
  - S51-S60 (Secondary): Opportunistic users. The BS updates LAST _ PRIMARY _ ACTIVITY on receipt of frame from a Primary ID.

  Any attempt to transmit by a Secondary node within PRIMARY _ PROTECTION _ WINDOW (10s) is placed into a deque (Queue) and the BS attempts delivery again once the window has expired.

- Client-Side Listen Before Talk (LBT): The Dynamic _ Sender implements either an autonomous agent or a manual sender. It checks channel _ state _ BSx.json file before transmitting [14]
  - If Status = BUSY: This node then goes into a time.sleep () backoff loop.
  - If Status is IDLE: That node constructs the packet, encrypts with AES.MODE _ GCM, and sends.

- Data Visualization: Results are exported to PDF via matplotlib.backends.backend_pdf. This includes:
  - Constellation Plots: Visualization of the density of the QAM modulation.
  - OFDM I/Q Plots: Diagram of the time-domain signal structure.
  - Energy Timelines: A temporal graph with Primary User activity highlighted in red, showing the Protection Window logic.

5. Appendices

A. Simulation Parameters

- Carrier Frequency: Baseband simulation (centered at 0 Hz equivalent).
- Modulation Schemes: 16-QAM, 64-QAM, 256-QAM (Adaptive).
- Subcarriers (N_c): 16, 32, 64, 128 (Adaptive based on latency/mobility cost).
- Error Correction: Reed-Solomon (Codec size 40).
- Encryption: AES-256 in GCM Mode.

B. Node ID Mapping

The following ID ranges are strictly enforced by the Base Station logic:

- S01 - S10: JAZZ (Primary)
- S11 - S20: WARID (Primary)
- S21 - S30: UFONE (Primary)
- S31 - S40: TELENOR (Primary)
- S41 - S50: ZONG (Primary)
- S51 - S60: SECONDARY (Opportunistic Access)

C. Control Protocol Commands

The system uses the following string literals for connection management:

- __CONNECT_REQUEST__: Initiate handshake.
- __CONNECT_ACCEPT__: Establish session.
- __CONNECT_REJECT__: Deny session.
- __DISCONNECT__: Terminate session.

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